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"Out of the Loop": Autonomous Weapon Systems and the Law of Armed Conflict

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I. Introduction

The introduction of autonomous weapon systems into the "battlespace" will profoundly influence the nature of future warfare. This reality has begun to draw the attention of the international legal community. Most notably, in November 2012, Human Rights Watch released *Losing Humanity: The Case against Killer Robots*.[1] The report is a scathing critique of autonomous weapons systems and has sparked a lively and important debate over the lawfulness of taking humans "out of the loop" during lethal targeting.[2]

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[1] HUMAN RIGHTS WATCH, LOSING HUMANITY: THE CASE AGAINST KILLER ROBOTS (2012), *available at* http://www.hrw.org/sites/default/files/reports/arms1112ForUpload_0_0.pdf [hereinafter LOSING HUMANITY].

[2] On the issue of the legality of autonomous weapon systems, see Jeffrey S. Thurnher, *The Law that Applies to Autonomous Weapon Systems*, 17 ASIL INSIGHTS 4 (Jan. 18, 2013), http://www.asil.org/pdfs/insights/insight130118.pdf; Kenneth Anderson & Matthew Waxman, *Law and Ethics for Robot Soldiers*, 176 POL. REV. (Dec. 1, 2012), http://www.hoover.org/publications/policy-review/article/135336; Markus Wagner, *Taking Humans Out of the Loop: Implications for International Humanitarian Law*, 21 J. L. INFO. &

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Human Rights Watch's position on "robots," a colloquial rendering for autonomous weapon systems, is forceful and unambiguous: "[F]ully autonomous weapons would not only be unable to meet legal standards but would also undermine essential non-legal safeguards for civilians."[3] The organization accordingly concludes that they "should be banned and . . . governments should urgently pursue that end."[4]

This conclusion has important national security implications. The United States and its allies have a substantial interest in maintaining a technological edge over potential adversaries, in particular by fielding systems that enable them to deliver lethal force while minimizing the risk to their own forces. After all, the objective of warfare is to achieve and exploit advantages to the enemy's detriment. Sensitive to this reality, the law of armed conflict has never been about ensuring a "fair fight"; rather, it comprises prohibitions, restrictions, and obligations designed to balance a State's interest in effectively prosecuting the war (military necessity) with its interest in minimizing harm to those involved in a conflict (humanity).[5] The question for the legal community, therefore, is whether autonomous weapon systems comply with the legal norms that States have put in place to achieve this balance. Any misinterpretation or misapplication of said norms will necessarily skew this delicate balance, thereby placing compliance by States with the law of armed conflict at risk.

Furthermore, as has too often been the case with international calls to ban particular weapons, debates over weapons have had a way of getting off-course, both normatively and operationally. At least from the United States government's perspective, this tendency contributed to its unwillingness to become Party to the Ottawa Antipersonnel Land Mines and the Dublin Cluster Munitions Conventions.[6] A contemporary example

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SCI. 155, 155 (2011), *available at* http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1874039.

[3] LOSING HUMANITY, *supra* note 1, at 1–2.

[4] *Id.* at 2.

[5] *See generally* Michael N. Schmitt, *Military Necessity and Humanity in International Humanitarian Law: Preserving the Delicate Balance,* 50 VA. J. INT'L L. 795 (2010).

[6] Convention on the Prohibition of the Use, Stockpiling, Production and Transfer of Anti-Personnel Mines and on Their Destruction, Sept. 18, 1997, 2056 U.N.T.S. 211; Convention on Cluster Munitions, Dec. 3, 2008, 48 I.L.M. 357. Although the United States elected not to become Party to the Ottawa Convention, the Obama Administration is presently reviewing its position on the matter. Andrew J. Shapiro, *Briefing on the Release of*

presaging the emerging controversy over autonomous weapon systems is the polemic over drones, which has often been counter-normative and counter-factual.[7] Rather than allowing the discourse over autonomous weapon systems to similarly spiral out of control, what is needed is a vibrant, informed, measured, and mature discussion of the relevant legal issues.

This Article is intended to help infuse granularity and precision into the legal debates surrounding such weapon systems and their future uses. It suggests that whereas some conceivable autonomous weapon systems might be prohibited as a matter of law, the use of others will be unlawful only when employed in a manner that runs contrary to the law of armed conflict's prescriptive norms governing the "conduct of hostilities."[8] This Article concludes that an outright ban of autonomous weapon systems is insupportable as a matter of law, policy, and operational good sense. Indeed, proponents of a ban underestimate the extent to which the law of armed conflict, including its customary law aspect, will control autonomous weapon system operations.[9] Some autonomous weapon systems that might

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the Tenth Edition of the "To Walk the Earth in Safety" Report, U.S. DEP'T OF ST. (Dec. 19, 2011), http://www.state.gov/t/pm/rls/rm/179143.htm. For the U.S. position on cluster munitions, *see* U.S. DEP'T. OF STATE, CLUSTER MUNITIONS, http://www.state.gov/t/pm/wra/c25930.htm (last visited March 21, 2013).

[7] Michael N. Schmitt, *Unmanned Combat Aircraft Systems (Armed Drones) and International Humanitarian Law: Simplifying the Oft Benighted Debate,* 30 B.U. INT'L L. J. 595, 596 (2012).

[8] The phrase "conduct of hostilities" refers in particular to "attacks," as that term is understood in Article 49 of Additional Protocol I. Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts, art. 49, June 8, 1977, 1125 U.N.T.S. 3 [hereinafter Additional Protocol I]. The core conduct of hostility rules are set forth in Part IV, Section I, of the Protocol. Although the United States is not a Party to the Protocol, most of the rules contained in the Section reflect customary international law.

[9] We refer to the following non-binding compilations of customary international humanitarian law rules to draw conclusions as to customary status of the norms referenced in this article: INTERNATIONAL COMMITTEE OF THE RED CROSS, CUSTOMARY INTERNATIONAL HUMANITARIAN LAW (Jean-Marie Henckaerts & Louise Doswald-Beck eds., 2005) [hereinafter CUSTOMARY INTERNATIONAL HUMANITARIAN LAW STUDY]; MICHAEL N. SCHMITT, CHARLES H.B. GARRAWAY & YORAM DINSTEIN, THE MANUAL ON THE LAW OF NON-INTERNATIONAL ARMED CONFLICT WITH COMMENTARY (2006), *available at* http://www.iihl.org/iihl/Documents/The%20Manual%20on%20the%20Law%20of%20NIAC.pdf; PROGRAM ON HUMANITARIAN POLICY AND CONFLICT RESEARCH AT HARVARD UNIVERSITY (HPRC), COMMENTARY ON THE MANUAL ON INTERNATIONAL LAW APPLICABLE TO AIR AND MISSILE WARFARE (2010) [hereinafter AMW MANUAL COMMENTARY], TALLINN MANUAL ON THE INTERNATIONAL LAW APPLICABLE TO CYBER WARFARE (Michael N. Schmitt gen. ed., 2013 forthcoming) [hereinafter TALLINN

be developed would already be unlawful *per se* under existing customary law, irrespective of any treaty ban. The use of certain others would be severely limited by that law.

Furthermore, an outright ban is premature since no such weapons have even left the drawing board.[10] Critics typically either fail to take account of likely developments in autonomous weapon systems technology or base their analysis on unfounded assumptions about the nature of the systems. From a national security perspective, passing on the opportunity to develop these systems before they are fully understood would be irresponsible. Perhaps even more troubling is the prospect that banning autonomous weapon systems altogether based on speculation as to their future form could forfeit their potential use in a manner that would minimize harm to civilians and civilian objects when compared to non-autonomous weapon systems.

II. Understanding Autonomous Weapon Systems

Before turning to the law, it is essential to examine the operational and technological context in which that law will be applied. The starting point for any such discussion is definitional. A "weapon system" consists of a weapon and the items associated with its employment.[11] An example is an

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MANUAL]. Additionally, although they are not compilations of customary international law, scholars and practitioners often look to the Rome Statute and the U.S. Commander's Handbook as strong indications of a norm's customary status, the latter in light of the United States' Additional Protocol I non-Party status. Statute of the International Criminal Court, July 17, 1998, 2187 U.N.T.S. 90 [hereinafter Rome Statute]; U.S. NAVY/U.S. MARINE CORPS/U.S. COAST GUARD, THE COMMANDER'S HANDBOOK ON THE LAW OF NAVAL OPERATIONS, NWP 1-14M/MCWP 5-12.1/COMDTPUB P5800.7A (2007), *available at* http://www.usnwc.edu/getattachment/a9b8e92d-2c8d-4779-9925-0defea93325c/1-14M_(Jul_2007)_(NWP) [hereinafter COMMANDER'S HANDBOOK].

[10] The International Committee of the Red Cross sagely suggests taking time to analyze carefully the legal and other issues surrounding autonomous weapon systems before drawing conclusions as to their lawfulness. INT'L COMM. OF THE RED CROSS, INTERNATIONAL HUMANITARIAN LAW AND THE CHALLENGES OF CONTEMPORARY ARMED CONFLICTS 40 (Report of the 31st Conference of the Red Cross and Red Crescent, Geneva, Switzerland 31IC/11/5.1.2 2011) , *available at* http://www.icrc.org/eng/assets/files/red-cross-crescent-movement/31st-international-conference/31-int-conference-ihl-challenges-report-11-5-1-2-en.pdf [hereinafter ICRC REPORT].

[11] In this Article, we use the term "weapon system" as shorthand to refer both to a weapon and a complete weapon system.

F-16 fighter-bomber armed with a Joint Direct Attack Munition (JDAM) bomb.

The Department of Defense (DoD) has defined an *autonomous* weapon system as:

> A weapon system that, once activated, can select and engage targets without further intervention by a human operator. This includes human-supervised autonomous weapon systems that are designed to allow human operators to override operation of the weapon system, but can select and engage targets without further human input after activation.[12]

The crux of full autonomy, therefore, is the capability to identify, target, and attack a person or object without human interface. Although a human operator may retain the ability to take control of the system, it is capable of operating on its own. Of course, a fully autonomous system is never entirely human-free. Either the system designer or an operator would at least have to program it to function pursuant to specified parameters, and an operator would have to decide to employ it in a particular battlespace.

Fully autonomous weapon systems must be distinguished from other categories of weapons with lesser degrees of autonomy, which Human Rights Watch has not suggested banning. For instance, U.S. forces have operated two "human-supervised" autonomous systems for many years—the Aegis at sea and the Patriot on land—both designed to defend against short notice missile attacks.[13] Another human-supervised

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[12] U.S. DEP'T OF DEF., Directive 3000.09, AUTONOMY IN WEAPON SYSTEMS 13–14 (Nov. 2, 2012), *available at* http://www.dtic.mil/whs/directives/corres/pdf/300009p.pdf [hereinafter DOD DIRECTIVE 3000.09]. Human Rights Watch distinguishes three categories of systems. A "human-*in*-the-loop" system requires a human to direct the system to select a target and attack it. The Department of Defense labels these "semi-autonomous systems." A "human-*on*-the-loop" weapon is one in which the system selects targets and attacks them, albeit with human operator oversight. The Department of Defense term is "human-supervised autonomous system." Finally, Human Rights Watch calls a system that can attack without any human interface a "human-*out-of*-the-loop weapon." The Department of Defense moniker is "fully autonomous weapon system." LOSING HUMANITY, *supra* note 1, at 2.

[13] Lockheed Martin, *Aegis Combat System*, http://www.lockheedmartin.com/us/products/aegis.html (last visited Feb. 23, 2013); Federation of American Scientists (FAS), *Patriot TMD*, http://www.fas.org/spp/starwars/program/patriot.htm (last visited Feb. 23, 2013).